

Optimization 1

ISE 5406

Lecture 8: Gradient Monotonicity, Optimal Solutions & Quasiconvexity

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 7): Duality Theory

- The Lagrangian dual problem and dual function
- Weak duality theorem: $d^* \leq p^*$
- Strong duality and Slater's condition
- Economic interpretation: dual variables as shadow prices

Today (Lecture 8): Gradient Monotonicity, Optimal Solutions & Quasiconvexity

- **Part I — Differentiable Convex Functions:**
 - Gradient monotonicity: a third characterization of convexity
 - Characterization of optimal solution sets
 - Convex maximization at extreme points
- **Part II — Beyond Convexity:**
 - Quasiconvex functions and level set characterization
 - Strictly quasiconvex functions and quasiconvex maximization

Next Time (Lecture 9): Convex and Quasiconvex Maximization

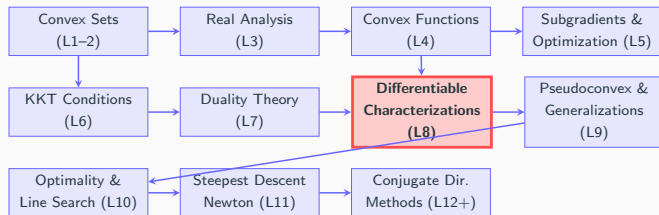
- More on maximization over polyhedral sets
- Pseudoconvex functions and their properties
- Relationships between generalized convexity classes

Coming Soon (Lectures 10–11): Algorithms and Convergence

- Gradient descent and step size selection
- Convergence rates for convex and strongly convex functions
- From theory to computation

Big Picture: Building on the first- and second-order characterizations from Lecture 4, today we develop gradient monotonicity as a new convexity test, characterize optimal solution sets, and broaden our view to quasiconvex functions, which preserve many useful optimization properties with weaker assumptions.

Big Picture: Course Roadmap



Why Gradient Monotonicity & Generalizations Matter

- Gradient monotonicity $\Rightarrow$ a new, practical convexity test via gradient comparisons
- Optimal solution set characterization $\Rightarrow$ understanding solution structure
- Quasiconvex functions broaden the class of tractable optimization problems
- Extreme point solutions for maximization $\Rightarrow$ connects to LP theory

Reading for This Lecture

Textbook References

- Sections 3.3, 3.4, and 3.5 of Bazaraa, Sherali, and Shetty (2006)
- Section 3.4 of Boyd and Vandenberghe (Convex Optimization)
- **Supplementary:** Sections 2.1–2.3 of Wright and Recht (2022) — differentiable convex functions from a data science perspective

Key Topics

- Gradient monotonicity — a new characterization of convexity
- Optimal solution sets for convex minimization
- Convex maximization at extreme points
- Quasiconvex and strictly quasiconvex functions

Outline

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Recap: Key Results from Lecture 4

We proved the following in Lecture 4. These are the foundation for today's new results.

First-Order Characterization (Lecture 4)

f differentiable on open convex S is convex $\Leftrightarrow$ for all $\mathbf{x}, \bar{\mathbf{x}} \in S$:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}).$$

Interpretation: tangent hyperplanes are global underestimators.

Second-Order Characterization (Lecture 4)

f twice differentiable on open convex S is convex $\Leftrightarrow \nabla^2 f(\bar{\mathbf{x}}) \succeq 0$ for all $\bar{\mathbf{x}} \in S$.

Practical test: compute Hessian and check positive semidefiniteness.

Today: We develop a *third* characterization — **gradient monotonicity** — and then move beyond convexity to broader function classes (quasiconvex, strictly quasiconvex) that preserve key optimization properties.

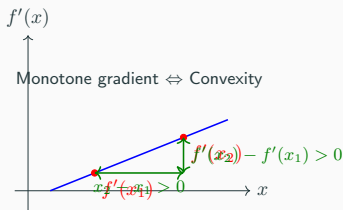
Gradient Monotonicity Characterization

The result below provides another necessary and sufficient characterization of differentiable convex functions.

Theorem: Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set and $f : S \rightarrow \mathbb{R}$ be differentiable. Then f is convex if and only if for each $\mathbf{x}_1, \mathbf{x}_2 \in S$:

$$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0. \quad (1)$$

Note: For univariate functions, (1) implies the slope is nondecreasing.



Proof of Gradient Monotonicity

Theorem (for reference)

f differentiable on open convex S is convex $\Leftrightarrow$

$$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \text{ for all } \mathbf{x}_1, \mathbf{x}_2 \in S.$$

Proof Outline:

1. ($\Rightarrow$) Assume f is convex. By the first-order condition:

$$f(\mathbf{x}_1) \geq f(\mathbf{x}_2) + \nabla f(\mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2)$$

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)$$

Adding these two inequalities:

$$0 \geq [\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_1 - \mathbf{x}_2)$$

which gives $[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$. $\checkmark$

2. ($\Leftarrow$) Uses the **Mean Value Theorem**: for $\mathbf{x}_1, \mathbf{x}_2 \in S$, there exists $\lambda \in (0, 1)$ such that:

$$f(\mathbf{x}_2) - f(\mathbf{x}_1) = \nabla f(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2)^T(\mathbf{x}_2 - \mathbf{x}_1)$$

Apply gradient monotonicity to show the first-order condition holds.

$\checkmark$

See instructor proof notes for complete details.

Why Gradient Monotonicity? Practical Applications

Why do we need a third characterization? The first- and second-order conditions from Lecture 4 are powerful, but gradient monotonicity fills practical gaps:

Application 1: Data-Driven Convexity Verification

In machine learning, you often have access to gradient evaluations $\nabla f(\mathbf{x}_i)$ at sample points but computing the full Hessian is prohibitively expensive (e.g., deep networks with millions of parameters). Gradient monotonicity lets you **test convexity from gradient data alone**: if $[\nabla f(\mathbf{x}_i) - \nabla f(\mathbf{x}_j)]^T (\mathbf{x}_i - \mathbf{x}_j) < 0$ for *any* pair, the loss is not convex.

Application 2: Monotone Operator Theory

Gradient monotonicity is the bridge to **variational inequalities** and **monotone operator theory**, which underpin modern algorithms like the *extragradient method*, *forward-backward splitting*, and *ADMM*. When ∇f is monotone, the operator $T = \nabla f$ defines a variational inequality whose solutions correspond to minimizers of f .

Application 3: Economics — Revealed Preference

In economics, gradient monotonicity of a cost function implies that marginal costs increase with output — the “law of diminishing returns” can be verified by comparing marginal costs at different production levels, without knowing the full cost surface.

Exercise 1: Gradient Monotonicity

Think-Pair-Share

Consider the functions below. Use gradient monotonicity to determine convexity.

1. $f(x) = e^x$ on $\mathbb{R}$. Verify that $(f'(x_2) - f'(x_1))(x_2 - x_1) \geq 0$ for $x_1 = 0, x_2 = 1$.
2. $f(x) = x^3$ on $\mathbb{R}$. Check gradient monotonicity for $x_1 = -1, x_2 = 0$. What does this tell you about convexity of f ?
3. $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2$. Check the condition for $\mathbf{x}_1 = (1, 0)^T$ and $\mathbf{x}_2 = (0, 1)^T$. What does this tell you?

Hint: A single violation of the inequality is enough to disprove convexity.

Solution: Gradient Monotonicity

Answers

(1) $f'(x) = e^x$. At $x_1 = 0$: $f'(0) = 1$. At $x_2 = 1$: $f'(1) = e$.

$(e - 1)(1 - 0) = e - 1 \approx 1.72 > 0$. Gradient monotonicity holds. ✓

In fact, $f''(x) = e^x > 0$ everywhere, confirming f is convex on all of $\mathbb{R}$.

(2) $f'(x) = 3x^2$. At $x_1 = -1$: $f'(-1) = 3$. At $x_2 = 0$: $f'(0) = 0$.

$(0 - 3)(0 - (-1)) = (-3)(1) = -3 < 0$. Gradient monotonicity fails!

Therefore f is NOT convex on $\mathbb{R}$. The gradient is *decreasing* on $(-\infty, 0)$, which gradient monotonicity detects directly — no need to check the Hessian.

(3) $\nabla f = (x_2, x_1)^T$.

$\nabla f(\mathbf{x}_1) = (0, 1)^T$, $\nabla f(\mathbf{x}_2) = (1, 0)^T$.

$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T (\mathbf{x}_2 - \mathbf{x}_1) = (1, -1)^T \cdot (-1, 1)^T = -1 - 1 = -2 < 0$.

Gradient monotonicity fails! Therefore f is NOT convex on $\mathbb{R}^2$.

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Optimal Solution Set Characterization

Note: $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in S} f(\mathbf{x})$ means $\mathbf{x}^*$ is an optimal solution to $\min_{\mathbf{x} \in S} f(\mathbf{x})$.

Theorem

Let $S \subseteq \mathbb{R}^n$ be a nonempty convex set and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a twice differentiable convex function. If $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in S} f(\mathbf{x})$, then the set of optimal solutions can be characterized as:

$$\arg \min_{\mathbf{x} \in S} f(\mathbf{x}) =: S^* = \{\mathbf{x} \in S : \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \leq 0, \nabla f(\mathbf{x}^*) = \nabla f(\mathbf{x})\}.$$

Interpretation: The optimal set consists of all feasible points that:

1. Do not increase the objective (gradient condition:
 $\nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \leq 0$), and
2. Have the same gradient as $\mathbf{x}^*$ (flat region condition:
 $\nabla f(\mathbf{x}^*) = \nabla f(\mathbf{x})$)

Why We Care: Optimal Sets in Practice

Understanding the *structure* of the optimal set — not just finding one optimum — matters in many applications:

Machine Learning: The “Flat Valley” Phenomenon

In overparametrized models (e.g., deep neural networks), the loss surface often has large, flat optimal sets. The theorem’s “constant gradient” condition ($\nabla f(\mathbf{x}^*) = \nabla f(\mathbf{x})$ on S^*) explains why: all optimal solutions share the same gradient landscape. Research shows that wider optimal sets lead to **better generalization** (Keskar et al., 2017).

Portfolio Optimization: Multiple Equally-Optimal Allocations

When minimizing portfolio risk $(x_1 + x_2 - \text{target})^2$ over a budget constraint, the optimal set is often a *face* of the feasible polytope — meaning many allocations achieve the same risk. An investor can choose among these based on secondary preferences (liquidity, diversification) while maintaining optimality.

Sensitivity Analysis & Robustness

The “polyhedral optimal set” corollary tells us that for convex quadratic objectives over polytopes, S^* has a clean structure. This is critical for **post-optimality analysis**: understanding how the optimal set changes with small perturbations to the data.

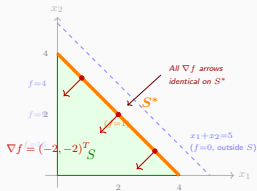
Geometric Interpretation of the Optimal Set

Concrete Example: $f(x_1, x_2) = (x_1 + x_2 - 5)^2$ over

$$S = \{x \in \mathbb{R}_+^2 : x_1 + x_2 \leq 4\}.$$

Convex but *not* strictly convex ($H = 2\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq 0$, rank 1). Unconstrained min:

$f = 0$ on line $x_1 + x_2 = 5$ (outside S).



Check: $\nabla f(x) = 2(x_1 + x_2 - 5)(1, 1)^T$. On S^* ($x_1 + x_2 = 4$): $\nabla f \equiv (-2, -2)^T$ — equal gradients ✓.

$\nabla f(x^*)^T(x - x^*) = -2((x_1 + x_2) - 4) \leq 0$ for $x \in S^*$ (equality), and > 0 for interior points ✓

Key Corollaries

- **Corollary 1:**

$$S^* = \{x \in S : \nabla f(x^*)^T(x - x^*) = 0 \text{ and } \nabla f(x^*) = \nabla f(x)\}$$

- **Corollary 2:** If S is polyhedral and f is a convex quadratic, then S^* is polyhedral

Exercise 2: Characterizing Optimal Solutions

Group Exercise

Consider $f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 1)^2$ over
 $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 \leq 2, \mathbf{x} \geq \mathbf{0}\}$.

1. Is f convex? What is $\nabla f(\mathbf{x})$?
2. Find the unconstrained minimizer. Is it in S ?
3. Find the constrained optimum $\mathbf{x}^*$ and characterize S^* .
4. Is S^* a singleton? Why or why not?

Hint: The unconstrained minimizer is $(2, 1)^T$. Check if it satisfies $x_1 + x_2 \leq 2$.

Solution: Characterizing Optimal Solutions

Answers

(1) $\nabla^2 f = 2I \succ 0 \Rightarrow$ strictly convex.

$$\nabla f(\mathbf{x}) = (2(x_1 - 2), 2(x_2 - 1))^T.$$

(2) Unconstrained: $\nabla f = 0 \Rightarrow \mathbf{x} = (2, 1)^T$. Check: $2 + 1 = 3 > 2$.

Not in S .

(3) The constraint $x_1 + x_2 \leq 2$ must be active. Minimize

$$(x_1 - 2)^2 + (x_2 - 1)^2 \text{ subject to } x_1 + x_2 = 2, \mathbf{x} \geq 0.$$

Substituting $x_2 = 2 - x_1$: minimize

$$(x_1 - 2)^2 + (1 - x_1)^2 = 2x_1^2 - 6x_1 + 5.$$

Setting derivative to zero: $4x_1 - 6 = 0 \Rightarrow x_1 = 3/2, x_2 = 1/2$.

$\mathbf{x}^* = (3/2, 1/2)^T$ is feasible. $f(\mathbf{x}^*) = 1/4 + 1/4 = 1/2$.

(4) Yes, S^* is a singleton (unique optimum) because f is strictly convex. Strictly convex functions have at most one minimizer over any convex set.

Maximizing a Convex Function

Theorem: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. If $\mathbf{x}^* \in S$ is a local maximum to $\max_{\mathbf{x} \in S} f(\mathbf{x})$, then

$$\xi^T(\mathbf{x} - \mathbf{x}^*) \leq 0 \text{ for all } \mathbf{x} \in S,$$

where ξ is any subgradient of f at $\mathbf{x}^*$.

Corollary: If f is also differentiable and $\mathbf{x}^* \in S$ is a local maximum, then $\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \leq 0$ for all $\mathbf{x} \in S$.

Note: These conditions are **necessary but not sufficient** for optimality.

Example: $f(x) = x^2$, $S = [-1, 2]$. At $\bar{x} = 0$: $f'(0) = 0$, so

$0 \cdot (x - 0) = 0 \leq 0$ for all x . But $\bar{x} = 0$ is a *minimum*, not a maximum!

The maximum is at $x = 2$.

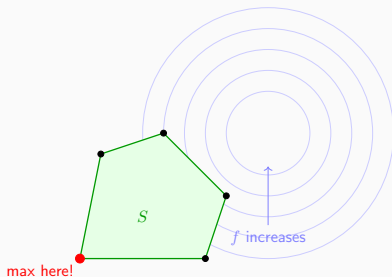
Convex Maximization Has an Extreme Point Solution

Theorem

Let S be a nonempty compact polyhedral set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. The problem $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution that is an extreme point of S .

This is a generalization of the fundamental theorem of LP!

Can be generalized to non-polyhedral compact convex sets S .



Intuition: A convex function “pushes outward,” so maximizing it over a polytope always lands at a vertex.

Convex Maximization in Practice

Maximizing a convex function over a polytope arises surprisingly often:

Robust Optimization: Worst-Case Analysis

In robust engineering design, you often need to find the **worst-case stress or load** across a feasible operating range (a polytope). The stress function is typically convex, so the extreme-point theorem tells you: *check only the vertices!* This reduces an infinite search to finitely many evaluations.

Experimental Design: D-Optimality

In statistics, **D-optimal design** maximizes $\det(X^T X)$ (the information matrix determinant) over a design polytope. Since $\log \det$ is concave, $-\log \det$ is convex, and maximizing $\det$ is equivalent to minimizing a convex function's negative. The theory implies optimal designs occur at the extreme points of the design space — i.e., at the boundary of the experimental range.

Combinatorial Optimization Connection

The extreme-point result generalizes the **fundamental theorem of LP** (optimal solution at a vertex). This is why LP relaxations of integer programs are so powerful: even when we relax integrality, the convex structure pushes solutions toward vertices, which are often integral.

Exercise 3: Convex Maximization

Group Exercise

Let $f(\mathbf{x}) = (x_1 - \frac{3}{2})^2 + (x_2 - 5)^2$ and

$$S = \{\mathbf{x} \in \mathbb{R}_+^2 : -x_1 + x_2 \leq 2, 2x_1 + 3x_2 \leq 11\}.$$

1. Verify that f is convex.
2. What is $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$? Where is the minimizer?
3. What is $\max\{f(\mathbf{x}) : \mathbf{x} \in S\}$? Which extreme point achieves it?

Hint: The extreme points of S are $(0,0)$, $(11/2,0)$, $(1,3)$, $(0,2)$. Evaluate f at each.

Solution: Convex Maximization

Answers

(1) $\nabla^2 f = 2I \succ 0$. Strictly convex. ✓

(2) Unconstrained minimizer: $\mathbf{x} = (3/2, 5)^T$. Check feasibility:
 $-(3/2) + 5 = 7/2 > 2$ and $2(3/2) + 3(5) = 18 > 11$. Not feasible.

Need to solve the constrained problem (e.g., via KKT). The minimum is at $\mathbf{x}^* = (1, 3)^T$ with $f(1, 3) = 1/4 + 4 = 17/4 = 4.25$.

(3) Evaluate f at extreme points:

- $f(0, 0) = 9/4 + 25 = 109/4 = 27.25$
- $f(11/2, 0) = (11/2 - 3/2)^2 + 25 = 16 + 25 = 41$
- $f(1, 3) = 1/4 + 4 = 17/4 = 4.25$
- $f(0, 2) = 9/4 + 9 = 45/4 = 11.25$

Maximum is $f(11/2, 0) = 41$ at the extreme point $(11/2, 0)$. ✓

Consistent with the theorem: max of a convex function occurs at an extreme point.

Proof: Extreme Point Solution for Convex Max

Theorem (Convex Maximization at Extreme Points)

Let S be a nonempty compact polyhedral set and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex.

Then $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution at an extreme point of S .

Proof Outline: (to be done on board)

1. Since S is compact and f is continuous, the maximum is attained at some $\mathbf{x}^* \in S$
2. If $\mathbf{x}^*$ is not an extreme point, write $\mathbf{x}^* = \sum_{i=1}^k \lambda_i \mathbf{x}_i$ as a convex combination of extreme points (by Minkowski's theorem for polytopes)
3. By convexity: $f(\mathbf{x}^*) = f(\sum \lambda_i \mathbf{x}_i) \leq \sum \lambda_i f(\mathbf{x}_i) \leq \max_i f(\mathbf{x}_i)$
4. But $f(\mathbf{x}^*)$ is the maximum over $S \Rightarrow$ equality must hold
5. Therefore $\max_i f(\mathbf{x}_i) = f(\mathbf{x}^*)$, and the achieving extreme point $\mathbf{x}_i$ is also optimal

See instructor proof notes for complete details.

Transition: Why Go Beyond Convexity?

So far in Part I, we developed powerful tools for differentiable convex functions:

- Three equivalent characterizations: tangent underestimators, positive semidefinite Hessian, gradient monotonicity
- Optimal solution sets are convex with constant gradient
- Maximization over polytopes always achieves at an extreme point

But convexity is a strong assumption!

Many important functions are *not* convex but still have useful structure:

- $f(x) = x^3$ has convex lower level sets even though f is not convex
- Ratios of linear functions (common in economics) are not convex
- Log-likelihood functions in statistics are often not convex

In Part II, we study **quasiconvex functions** — a broader class that relaxes the chord condition to a level-set condition, while preserving key optimization properties (e.g., maximization at extreme points).

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Quasiconvex Functions: Definition

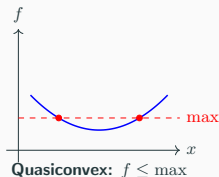
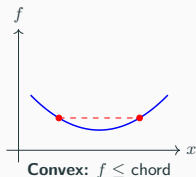
Definition

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is **quasiconvex** if for each $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}.$$

f is **quasiconcave** if $-f$ is quasiconvex.

Comparison with Convexity:



Key insight: Convexity says the function lies below the *chord*; quasiconvexity only requires it to lie below the *maximum* of the endpoints. Quasiconvexity is a weaker condition.

Quasiconvex Functions in the Wild

Quasiconvex functions are everywhere in applied optimization, even when convexity fails:

Economics: Utility Functions and Consumer Theory

Economists model consumer preferences with **quasiconcave** utility

functions (e.g., Cobb–Douglas: $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$). The upper level sets $\{\mathbf{x} : u(\mathbf{x}) \geq \alpha\}$ are the familiar *convex indifference curves*. Consumer demand theory relies entirely on quasiconcavity, not full concavity.

Signal Processing: Signal-to-Noise Ratios

Ratios like $\text{SNR}(\mathbf{x}) = \frac{\|\mathbf{a}^T \mathbf{x}\|^2}{\mathbf{x}^T \Sigma \mathbf{x}}$ are quasiconcave (the upper level sets are ellipsoids). Beamforming in antenna arrays maximizes such ratios over a constraint polytope — the extreme-point theorem applies.

Operations Research: Location Problems

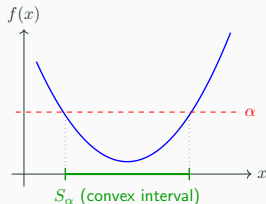
Distance-based objectives like $f(\mathbf{x}) = \max_i \|\mathbf{x} - \mathbf{a}_i\|$ (minimax facility location) are *quasiconvex*. The level sets $\{\mathbf{x} : \max_i \|\mathbf{x} - \mathbf{a}_i\| \leq r\}$ are intersections of balls, hence convex. This enables bisection algorithms on the radius parameter r .

Common thread: In each case, the function isn't convex, but the *level sets* are — and that's enough for powerful optimization guarantees.

Level Set Characterization of Quasiconvexity

Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is quasiconvex if and only if the lower level set $S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ is convex for each $\alpha \in \mathbb{R}$.



Examples:

(a) $f(x) = x^3$ is quasiconvex (level sets $\{x : x^3 \leq \alpha\} = (-\infty, \alpha^{1/3}]$ are convex)

(b) $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is quasiconcave (upper level sets are convex intervals)

(c) Every convex function is also quasiconvex (but not vice versa!)

Proof: Level Set Characterization of Quasiconvexity

Theorem

f is quasiconvex on convex $S \Leftrightarrow S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ is convex for each $\alpha \in \mathbb{R}$.

Proof: (to be done on board)

1. $(\Rightarrow)$ Assume f is quasiconvex. Take $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$.
2. Then $f(\mathbf{x}_1) \leq \alpha$ and $f(\mathbf{x}_2) \leq \alpha$.
3. For $\lambda \in (0, 1)$: $f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\} \leq \alpha$.
4. So $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in S_\alpha \Rightarrow S_\alpha$ is convex. $\checkmark$
5. $(\Leftarrow)$ Assume all level sets are convex. Take $\mathbf{x}_1, \mathbf{x}_2 \in S$.
6. Let $\alpha = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$. Then $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$.
7. Since S_α is convex: $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in S_\alpha$
8. Therefore $f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \alpha = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$. $\checkmark$

See instructor proof notes for complete details.

Exercise 4: Quasiconvex Classification

Classification Challenge

For each function, determine if it is: (i) convex, (ii) quasiconvex, (iii) neither.

Function	Convex?	Quasiconvex?	Reasoning
(a) $f(x) = x^3$ on $\mathbb{R}$			
(b) $f(x) = \sqrt{x}$ on $\mathbb{R}_+$			
(c) $f(x) = \sin(x)$ on $[0, 2\pi]$			
(d) $f(x) = \max(x_1, x_2)$			
(e) $f(x) = x $ on $\mathbb{R}$			

Hint: Check level set convexity. A function is quasiconvex iff all lower level sets are convex.

Solution: Quasiconvex Classification

Answers

(a) $f(x) = x^3$ on $\mathbb{R}$: Not convex ($f''(x) = 6x$ changes sign).

Quasiconvex: $S_\alpha = (-\infty, \alpha^{1/3}]$ is convex for all α . (Function is monotone increasing.)

(b) $f(x) = \sqrt{x}$ on $\mathbb{R}_+$: Not convex ($f''(x) = -\frac{1}{4}x^{-3/2} < 0$; it's concave!). Quasiconvex and quasiconcave: monotone increasing $\Rightarrow$ both upper and lower level sets are intervals (hence convex).

(c) $f(x) = \sin(x)$ on $[0, 2\pi]$: Neither convex nor quasiconvex. The level set at $\alpha = 0.5$ is $\{x : \sin(x) \leq 0.5\} \approx [0, \pi/6] \cup [5\pi/6, 2\pi]$, which is not convex (not a single interval).

(d) $f(\mathbf{x}) = \max(x_1, x_2)$: Convex (pointwise max of affine functions) and hence quasiconvex.

(e) $f(x) = |x|$ on $\mathbb{R}$: Convex ($|x|$ is the max of x and $-x$, both affine) and hence quasiconvex.

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Properties and Pitfalls of Quasiconvex Functions

Important Notes on Quasiconvex Functions:

Warning 1: Local Optima $\neq$ Global Optima

A local optimal solution to a quasiconvex minimization problem is **not necessarily** a global optimal solution!

Warning 2: Not Closed Under Addition

A sum of quasiconvex functions is **not necessarily** quasiconvex!

Reference: Section 3.4.4 of Boyd and Vandenberghe covers operations that *do* preserve quasiconvexity (e.g., nonnegative weighted maximum, composition with monotone functions).

Key contrast with convexity:

Property	Convex	Quasiconvex
Local min = Global min	✓	×
Closed under addition	✓	×
Level sets convex	✓	✓
Closed under max	✓	✓

Strictly Quasiconvex Functions

Definition

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is **strictly quasiconvex** if for each $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $f(\mathbf{x}_1) \neq f(\mathbf{x}_2)$ and each $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}.$$

f is **strictly quasiconcave** if $-f$ is strictly quasiconvex.

Surprising fact: A strictly quasiconvex function is not necessarily quasiconvex!

Key Result

Every local minimum to a minimization problem with a *strictly* quasiconvex objective and a convex feasible region is a global minimum.

Why this matters: Strict quasiconvexity recovers the “local $\Rightarrow$ global” property that plain quasiconvexity lacks, making it useful for optimization.

Practical Impact

Many iterative algorithms (gradient descent, coordinate descent) can only guarantee convergence to *local* optima. Strict quasiconvexity is the weakest condition ensuring these local solutions are globally optimal — broader than convexity but with the same algorithmic guarantee. This is why identifying the function class matters before selecting an algorithm.

Quasiconvex Maximization: Extreme Points

Theorem

Let S be a nonempty compact polyhedral set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be quasiconvex and continuous on S . The problem $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution that is an extreme point of S .

Key insight: This generalizes the earlier result for convex maximization. Since every convex function is quasiconvex, this theorem strictly extends the class of functions for which “max at vertex” holds.

Hierarchy of extreme-point results:

1. **Linear functions (LP):** $\max \mathbf{c}^T \mathbf{x}$ over polytope $\Rightarrow$ vertex solution
2. **Convex functions:** $\max f(\mathbf{x})$ over polytope $\Rightarrow$ vertex solution
3. **Quasiconvex + continuous:** $\max f(\mathbf{x})$ over compact polytope $\Rightarrow$ vertex solution

Each level strictly generalizes the previous. The key assumption is the “max pushes to boundary” property inherited from quasiconvexity.

Exercise 5: Quasiconvex vs. Convex Maximization

Think-Pair-Share

1. Give an example of a quasiconvex function that is NOT convex.
2. Consider $f(x) = x^3$ on $S = [-1, 2]$.
 - Is f convex on S ? Is it quasiconvex on S ?
 - What is $\max_{x \in S} f(x)$? At which extreme point?
 - What is $\min_{x \in S} f(x)$? Is it at an extreme point?
3. True or False: If f is quasiconvex and we are minimizing f over a convex set, then every local minimum is a global minimum.

Solution: Quasiconvex vs. Convex Maximization

Answers

(1) $f(x) = x^3$ on $\mathbb{R}$. It is quasiconvex (monotone increasing $\Rightarrow$ all lower level sets are half-lines, hence convex) but NOT convex ($f''(x) = 6x$ changes sign).

Another example: $f(x) = \ln(x)$ on $(0, \infty)$ is concave (hence quasiconcave) and also quasiconvex (monotone), but not convex.

(2) On $S = [-1, 2]$:

- Not convex on $[-1, 2]$ (second derivative changes sign).
Quasiconvex (monotone increasing).
- $\max_{x \in [-1, 2]} x^3 = 8$ at $x = 2$ (extreme point). ✓
- $\min_{x \in [-1, 2]} x^3 = -1$ at $x = -1$ (also an extreme point, but this is because f is monotone).

(3) **False.** Quasiconvexity alone does not guarantee local min = global min. You need *strict* quasiconvexity for that property. A quasiconvex function can have flat regions that create non-global local minima.

Gradient Monotonicity — A Third Convexity Characterization

Gradient Monotonicity

Optimal Solutions & Convex Maximization

Optimal Solution Set Characterization

Convex Maximization at Extreme Points

Quasiconvex Functions: Definitions and Characterization

Strict Quasiconvexity & Quasiconvex Maximization

End-of-Lecture Assessment

Exercise 6: End-of-Lecture Assessment

Assessment: True/False and Short Answer

1. **True or False:** Gradient monotonicity
 $([\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$ for all $\mathbf{x}_1, \mathbf{x}_2$) is both necessary and sufficient for convexity of a differentiable function.
2. **True or False:** If the set of optimal solutions to $\min_{\mathbf{x} \in S} f(\mathbf{x})$ is nonempty (with f convex, S convex), then every convex combination of optimal solutions is also optimal.
3. **True or False:** Every convex function is quasiconvex, and every quasiconvex function is convex.
4. **Short Answer:** Why does the maximum of a convex function over a polytope always occur at a vertex?
5. **True or False:** If f is strictly quasiconvex, then every local minimum of $\min_{\mathbf{x} \in S} f(\mathbf{x})$ (with S convex) is a global minimum.

Solutions: End-of-Lecture Assessment

Answers

(1) **True.** This is the gradient monotonicity characterization proved in this lecture. For differentiable f on an open convex set, convexity $\Leftrightarrow$ gradient monotonicity. It complements the first- and second-order conditions from Lecture 4.

(2) **True.** The optimal solution set $X^* = \{\mathbf{x} \in S : f(\mathbf{x}) = f^*\}$ is convex when f and S are convex. If $\mathbf{x}_1, \mathbf{x}_2 \in X^*$, then $f(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) = f^*$, but also $\geq f^*$ since f^* is the minimum.

(3) **Half true, half false.** The first part is **true**: every convex function is quasiconvex (since $f \leq \text{chord} \leq \text{max of endpoints}$). The second part is **false**: $f(x) = x^3$ is quasiconvex but not convex.

(4) Because any interior point of a polytope can be written as a convex combination of vertices. By convexity of f :
 $f(\text{combination}) \leq \text{combination of } f(\text{vertices}) \leq \max f(\text{vertex})$. So a vertex achieves at least as large a value as any interior point.

(5) **True.** This is the key property of strict quasiconvexity. Unlike plain quasiconvexity, strict quasiconvexity ensures the “local $\Rightarrow$ global” property for minimization over convex sets.

Key Takeaways

Main Results

- **Gradient Monotonicity:** f convex $\Leftrightarrow (\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1))^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$ — a third characterization (alongside first- and second-order from Lecture 4)
- **Optimal Solution Set:** Characterized by gradient condition and constant-gradient property
- **Convex Max at Vertices:** $\max f$ over polytope always achieved at an extreme point
- **Quasiconvex $\Leftrightarrow$ Convex Level Sets:** Broader class of “well-behaved” functions
- **Strictly Quasiconvex:** Recovers local $\Rightarrow$ global property that quasiconvex lacks

Connections

- Gradient monotonicity $\rightarrow$ practical convexity verification via gradient comparisons
- Extreme point results $\rightarrow$ connects to LP theory and combinatorial optimization
- Quasiconvexity $\rightarrow$ level-set methods for broader problem classes
- Strict quasiconvexity $\rightarrow$ local $\Rightarrow$ global for minimization

Lecture 9 Preview: Convex and Quasiconvex Maximization

- Deeper study of maximization over polyhedral sets
- Pseudoconvex functions and their optimization properties
- Relationships between function classes: $\text{convex} \subset \text{pseudoconvex} \subset \text{strictly quasiconvex}$
- Practical implications for algorithm design

Big Picture: Today we established the analytical tools (characterizations, level sets, extreme points) that enable us to verify convexity and understand solution structure. Next lectures will build on these tools to develop and analyze optimization algorithms.

Questions?

Office hours: See syllabus for times. Feel free to ask about convexity characterizations, quasiconvex functions, or connections to your research!